ANDERSON SERANGOON JUNIOR COLLEGE

JC2 H2 Geography Preliminary Exam (2025)

H2 GEOGRAPHY **9173/02**

Paper 2 **2 September 2025**

3 hours

Additional Materials: Answer Booklet
1 Insert

READ THESE INSTRUCTIONS FIRST
If you have been given an Answer Booklet, follow the instructions on the front cover of the Booklet.
Write your name and class on the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use an HB pencil for any diagrams or graphs.
Do not use staples, paper clips, glue or correction fluid.

Answer Question 1 in **Section A**.
Answer Question 2 in **Section B**.
Answer **one** question in **Section C**.

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [ ] at the end of each question or part question.

Section A

Cluster 4: Fieldwork

1 Singapore's Islandwide Cycling Network (ICN) Programme aims to support the nation's car-lite vision and encourage greener, healthier commutes. Unveiled in 2020, there are currently more than 730km of cycling paths and park connectors across the island. By promoting cycling, the initiative demonstrates how sustainable transportation can play a vital role in strengthening community efforts to combat climate change.

A group of four Junior College students were tasked to investigate the community's views and levels of participation in sustainable transportation. They were given the one-week September school holidays to conduct their investigation. They decided to test the following hypothesis:

Young adults are more likely than the elderly to adopt cycling as a form of sustainable transportation in Singapore.

The students produced a questionnaire to collect information from a total of 40 residents at two different locations in Punggol. The students split themselves into pairs, with each pair conducting questionnaires for an hour at the respective locations from 12pm on a weekday of the September school holidays.

Resource 1 is a map showing the location of the two investigation sites in Punggol. Resource 2A shows a picture of investigation Site A. Resource 2B shows a picture of investigation Site B. Resource 3 is the questionnaire used by the students. Resource 4A shows the results by age group for question 3 of the questionnaire. Resource 4B shows the results by age group for question 4 of the questionnaire.

(a) Explain whether the hypothesis is suitable for the investigation. [3]

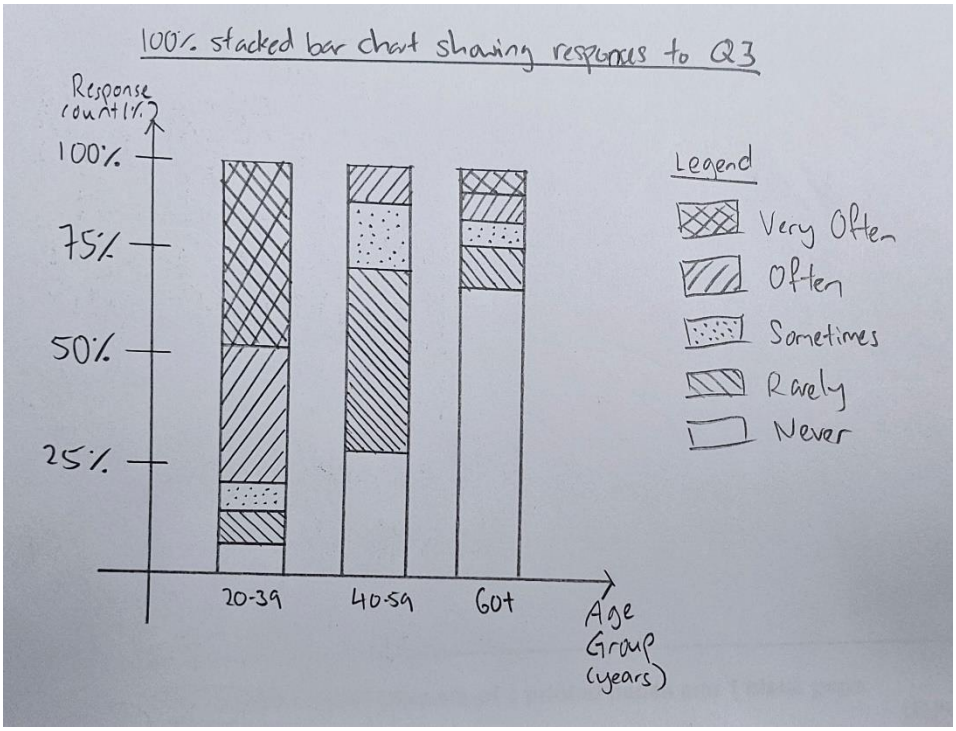
- The hypothesis is not at a suitable scale as Singapore is too large an area to investigate. There are only 4 students carrying out the investigation and pursuing the investigation in an area as large as Singapore will not be manageable for the students.
- The hypothesis is also not clearly defined as the specific age ranges of young adults and elderly are not identified. There are different definitions of the age range for young adults and elderly groups, and can lead to differing conclusions drawn from the collected data.
- However, the hypothesis is researchable as it is possible for students to collect data on the likelihood and opinions towards cycling through interviews and questionnaires.
- The hypothesis can also be considered clearly defined because it identified the specific target groups for the investigation – young adults and the elderly. It also clearly defines the specific aspect of sustainable transportation to be investigated, which is cycling.

(b) Suggest a suitable sampling method for the students to select participants for the questionnaire. [4]

- Stratified sampling is a suitable sampling method as it ensures the sample is more representative of the target groups and reduces sampling bias.

		• The students should conduct secondary research on census data for Punggol to determine the proportion for the number of respondents in each stratum – elderly and young adults. • For example, if census data shows that 25% of residents in Punggol are elderly and 15% of residents in Punggol are young adults, then the elderly stratum should have 25 respondents and the young adult stratum should have 15 respondents to make a total sample size of 40. • The students should then use a probability sampling method, such as systematic sampling, to collect the allocated number of responses from the relevant stratums. • The students can approach every n^{th} eligible person in the sampling site that walks past them until the quota for the stratum is filled, where n is a manageable fixed interval chosen by the students. • For example, if the students decide on n=3, then the students can approach every 3rd eligible young adult until 15 young adults have been surveyed.	
	(c)	With reference to Resources 1, 2A, and 2B, suggest how the students could minimise potential risks in conducting the survey.	[4]
		• Resource 1 shows that investigation site A is less than 1km away from where wild animals were recently sighted. Students should store food securely in sealed containers to prevent attracting the wild animals, and minimise the risk of being attacked by them. Students can also bring along a first aid kit in the event of being bitten or attacked by wild animals to minimise the risk of the damage worsening. • Resource 1 shows that Site B is located on challenging terrain. Students should wear shoes with good traction to minimise the risk of tripping and falling from the challenging terrain and being injured. • Resource 2A and 2B shows the presence of cycling paths where there might potentially be many cyclists on the road. Students should keep to the footpaths where possible, and have one student be on the lookout for oncoming cyclists to minimise the risk of colliding with cyclists and being injured. • Resource 2B shows the presence of a river within the investigation site. Students should keep a distance from the river, and wear shoes with good traction to minimise the risk of falling into the river and drowning or injuring themselves. • Resource 2A and 2B shows that the investigation site is not sheltered. Students should check the weather forecast and postpone the investigation in the event of inclement weather to minimise the risk of being struck by lightning.	
	(d)	With reference to Resources 1, 2A, and 2B, suggest how the students could minimise the impact of their investigation at the two sites.	[4]
		• Resource 1 shows that Site A is located about 1km away from residential areas. Students should keep their noise to a minimum to ensure that they do not disturb the residents living nearby. • Resource 1 also shows that Site A is located near to where wild animals were recently sighted. Students should stick to the pathways where possible	

		and avoid venturing into forested areas or approaching Lorong Halus Wetland to minimise disturbance to the wildlife. • Resource 2A and 2B shows that the cycling paths at the investigation sites are quite narrow. Students should stick to the footpaths so as to not obstruct cyclists who are cycling on the cycling path. • Resource 2A and 2B also shows that the footpaths at the investigation sites are very narrow. Students should avoid crowding in the middle of the footpath so as to not obstruct park users who are walking on the footpath. • Resource 2A and 2B also shows the presence of vegetation in close proximity to the pathways. Students should avoid stepping or trampling on the vegetation to preserve the integrity of the natural environment. • Resource 1 shows that both investigation sites are located within a Park Connector, which would be frequented by many park users. Students should ensure that they collect all litter and fieldwork materials from both sites before leaving to avoid polluting the area.	
	(e)	Explain how the design of the questionnaire shown in Resource 3 can be improved.	[6]
		• The questionnaire shown in Resource 3 requires the participants to indicate their names. However, this is not relevant to the investigation and can negatively impact the accuracy of data collected as participants might be less willing to provide truthful responses for fear of being identified. Therefore, participant names should not be required. • Question 2 in the questionnaire provides limited options for the common modes of transport, excluding potential modes such as personal mobility devices or wheelchairs, hindering the accuracy of data collected. Therefore, an 'others' option should be provided where participants can then indicate their most common mode of transport if it does not neatly fall into any of the listed options. • Question 3 in the questionnaire provides very vague and subjective options. For example, different participants may have different understandings of the terms 'rarely' and 'sometimes'. Instead, specific frequencies of cycling as a means of transport should be provided. For example, cycling once a week, cycling once a month, etc. • Question 4 provides limited options for challenges that discourage participants from cycling, hindering the accuracy of data collected as participants may have other challenges aside from the stated list. Instead, an 'others' options can be provided, or the question can be converted to an open-ended question to allow participants to elaborate on the challenges they might face. • Question 4 also has an assumption implied that participants face challenges from cycling, hindering accuracy of data collected. Instead, the questionnaire can be rephrased to 'What challenges might discourage people from cycling more often', replacing 'you' with 'people'. • Questions 5 and 6 has many technical jargons that participants may be unfamiliar with, such as 'modal distribution', 'transport mode share', 'carbon surplus', and 'carbon deficit'. These technical jargons should either be accompanied by a simple definition, or replaced with less technical, layman terms.	

	(f)	With the aid of a sketch diagram, explain how the students could better represent the data seen in Resource 4A	[5]
		 • A 100% stacked bar chart allows for clear visual comparison by making it easy to compare response patterns across the age groups at a glance, while a table of numbers is harder to interpret quickly. • Furthermore, a 100% stacked bar chart represents the data in proportions, highlighting the distribution of responses, which is more meaningful than raw counts. *Accept: Pie Charts with clear justification of how it is a better representation method*	
	(g)	Compare the results for question 4 for the 20-39 age group with the 60 and above age group shown in Resource 4B.	[4]
		• Both age groups face significant challenges relating to physical discomfort. This is evident from 7 out of 15 young adults, and 8 out of 15 elderly participants choosing 'Hot and humid weather' as a challenge. • However, the elderly face more significant challenges relating to health and safety compared to young adults. This is evident from the higher numbers of 7 out of 15 elderly choosing 'Lack of cycling paths / unsafe roads' and 10 out of 15 elderly choosing 'Health / physical limitations' as challenges, compared to the much lower numbers of 2 out of 15 for both challenges indicated by the young adults. • The young adults face more significant challenges relating to convenience compared to the elderly. This is evident from the higher numbers of 10 out of 15 young adults choosing 'Lack of bicycle parking' and 7 out of 15 young adults choosing 'Inconvenient for carrying belongings' as challenges, compared to the much lower numbers of 2 out of 15 and 1 out of 15 respectively for the elderly.	

	(h)	Evaluate the validity of the students' fieldwork at Punggol.	[10]
		Candidates can refer to Resources 4A and 4B and consider how much the results of the investigation supports the hypothesis. Candidates could refer to the Resources and consider how accurate and reliable the collected data are for drawing conclusions about the community's views and levels of participation in sustainable transportation. Candidates could look at the way data at the two sites are collected and assess if the methods used could make accuracy and reliability of the data questionable. Evaluation of these methods could include looking at the sample size, whether the samples selected are representative of the target population in Singapore, strengths and weaknesses of the different methods, etc. Additionally, candidates could consider what other data would be needed for this investigation. *Levels marked using Generic Level Descriptors for H2 fieldwork evaluative questions*	

Section B

Cluster 3: Sustainable Future and Climate Change

2	Resource 5 shows the spatial variations in ecological footprint across the world. Resource 6 shows waste generation and Gross Domestic Product (GDP) by country. Resource 7A shows the types of non-hazardous solid waste produced by income level. Resource 7B shows waste disposal methods by income level. Resource 8 shows a cartoon about a strategy to manage non-hazardous solid waste.		
	(a)	Describe the spatial variations in ecological footprint across the world as seen in Resource 5.	[4]
		• Generally, Northern Hemisphere has significantly higher ecological footprint compared to Southern Hemisphere • Specifically, continents in the NH such as North America and Europe has high EF of at least 4 gha/person, compared to continents in SH such as South America and Africa with EF of generally lower than planet biocapacity of 1.6 gha/person • However, some exceptions include: ○ Mexico, Central America & Caribbean island nations have relatively low EF of below 2.9 gha/person despite being in NH ○ Middle East generally has high EF ○ Australia, Chile, and Argentina have relatively high EF despite being in SH	
	(b)	Explain the difficulties faced when measuring sustainable urban development.	[4]
		<u>Deciding what aspects of dimensions to measure</u> • Sustainability has three broad dimensions (economic, social, environmental) which overlap and interact, making it hard to decide which aspects to prioritise. • Different cities face different challenges (e.g. air pollution vs. housing shortages), so what is "important" to measure varies by context. • Some aspects, like social wellbeing or cultural identity, are harder to define and quantify compared to physical ones like air quality. • Trade-offs occur: focusing on one aspect (e.g. green space) might neglect another (e.g. affordable housing). <u>Selecting appropriate indicators</u> • Indicators must be measurable, reliable, and comparable across cities, which is challenging as data availability varies. • Some indicators are too broad (e.g. "quality of life") while others are too narrow, failing to capture the full picture. • Long-term impacts of sustainability are difficult to capture with short-term indicators. • Different stakeholders (government, NGOs, citizens, businesses) may disagree on which indicators best represent sustainable progress.	
	(c)	Account for the pattern of waste generation shown in Resource 6.	[5]

		• Pattern: Positive relationship between waste generation and GDP per capita (i.e. the higher the GDP per capita, the higher the waste generated) <u>Explanation:</u> • Higher levels of consumption: Wealthier populations can afford to purchase more goods and services, which increases packaging waste, disposable items, and overall resource use. • Lifestyle and convenience culture: In high-income economies, busy urban lifestyles encourage reliance on ready-made meals, fast fashion, and single-use products, all of which produce more waste. • Shorter product life cycles: Rapid technological innovation and consumer demand for the latest ds to frequent disposal of still-usable items, increasing e-waste and other solid waste. • Urbanisation and industrialization: High GDP countries have more industries and services, producing commercial and industrial waste alongside household waste. Dense urban populations also centralise and magnify waste production. • Affluence enabling wasteful behaviour: Wealth reduces the need to economise; food waste is higher as people can afford to discard edible items, and replacement is cheaper than repair.	
	(d)	Contrast the types of non-hazardous solid waste produced and disposal methods between high- and low- income countries as seen in Resources 7A and 7B.	[6]
		<u>R7A</u> • Types of NHSW for High-income countries more evenly distributed (each type has at least 3.8%), while Low-income countries dominated by Food and Green and Others (2 types make up more than 80% of total) • High-income more varied types of NHSW compared to Low-income (which has no or insignificant contributions from rubber & leather, and wood) • Low-income have higher organic-to-inorganic waste type ratio compared to high-income countries <u>R7B</u> • HI more varied waste disposal methods, with 6 different types, while LI only has 4 different types, and dominated by open dump (comprising 93%) • Most common in HI is landfill, while most common in LI is open dump • More even distribution between waste disposal methods in HI compared to LI • HI has higher proportion of sustainable waste disposal methods compared to LI	
	(e)	Using Resource 8, suggest how the strategy to manage non-hazardous solid waste shown can impact sustainable urban development.	[5]
		<u>Economic</u> • Overspending on waste management reduces funds for housing, transport, and urban development, slowing economic growth → "allocate all of our city budget on expanding the landfill" • Valuable urban land is lost, reducing opportunities for commercial development and discouraging private investment → The Land Use Plan visually shows a huge portion of urban land consumed by landfills	

		<u>Social</u> • Residents' access to essential services, recreational spaces, and community facilities is reduced, lowering urban livability → "What about our houses and schools?" <u>Environment</u> • Large-scale landfills damage soil quality, reduce green space, and harm urban ecosystems, making the city less environmentally sustainable → The Land Use Plan and Landfill model visually shows a huge portion of urban land consumed by landfills • Prioritizing landfill expansion over sustainable infrastructure leads to increased pollution (air, water, soil), undermining long-term environmental health and resilience of the city. → "allocate all of our city budget on expanding the landfill"	
	(f)	Explain how urban population trends can hinder progress towards sustainable urban development.	[6]
		• High urbanisation rates and rapid urban growth o Pressure on infrastructure and services: Rapid growth often outpaces the provision of housing, transport, water, and waste systems. This reduces quality of life for current residents while creating infrastructure deficits that future generations will inherit, undermining SUD. o Environmental degradation: Expanding urban areas increase traffic emissions, loss of green space, and higher waste generation. This harms air quality and ecosystems today, while reducing the environmental resources available for future generations. o Social inequality: Migrants and the urban poor may be forced into informal settlements with limited access to education, healthcare, and jobs. This prevents inclusive development now and deepens long-term social divides, going against SUD's social dimension. o Resource strain: Demand for land, energy, and water rises sharply, leading to overconsumption. This compromises the ability of future generations to access sufficient resources, directly clashing with SUD principles. • Urban population loss o Economic decline: Population loss reduces tax revenue, leaving cities unable to maintain infrastructure or invest in sustainable projects like clean energy or efficient transport. This hinders current progress and weakens resilience for the future. o Vacant housing and derelict land: Abandoned buildings create urban blight and environmental hazards. This makes cities less liveable today and burdens future generations with costly regeneration needs, conflicting with SUD. o Ageing population pressures: Outmigration of younger workers leaves behind an ageing population. This raises dependency ratios and strains social services, reducing the capacity to sustain wellbeing for both current and future urban residents. o Inefficient land use: Maintaining oversized infrastructure for a smaller population wastes resources. This misallocation harms	

		present urban efficiency and locks future generations into unsustainable urban forms.	

Section C

Answer EITHER question 3 OR question 4.

3	'Issues faced by women living in the city pose the most significant hindrance to progress towards liveable cities.' Evaluate this statement.	[20]
	Possible Approaches: Candidates could approach the question by making a judgement on whether issues faced by women living in the city pose the most significant hindrance to progress towards liveable cities through a consideration of the range of issues faced by women living in cities. Candidates could evaluate the extent to which gendered issues such as safety, accessibility, affordability, and employment opportunities directly affect liveability. Candidates could also consider whether other groups (i.e. elderly) face issues that may be equally or more significant in hindering progress towards liveable cities. Candidates could also approach the question by making a judgement on whether issues faced by women living in the city are the most significant hindrance through a comparison of case studies. Candidates could evaluate whether gender-specific urban challenges are more impactful in some contexts than others. Candidates could also consider whether broader structural challenges (e.g. housing shortage, waste management issues, etc.) outweigh or exacerbate gendered issues in determining overall liveability. Candidates could also approach the question by making a judgement through a consideration of scale and temporality. Candidates could analyse whether the hindrances posed by women's issues are significant in the short term or long term, and whether they are most critical at the local, national, or global scale in affecting progress towards liveable cities. *Levels marked using Generic Level Descriptors for 20m H2 essays.*	
4	Evaluate the relative importance of natural and human factors in explaining contemporary climate change.	[20]
	Candidates could argue that human factors are overwhelmingly the dominant cause of contemporary climate change, particularly since the mid-20th century, rendering them significantly more important than natural factors. Their response must acknowledge that natural factors (like solar variability and volcanic eruptions) have always influenced climate, but modern scientific consensus—supported by evidence like the distinct isotopic fingerprint of fossil fuel carbon—shows they cannot account for the rapid and profound warming observed today. Instead, the rapid rise in anthropogenic greenhouse gas emissions (from burning fossil fuels, deforestation, and industrial agriculture) is the key driver, amplifying the natural greenhouse effect and thus establishing human activity as the primary explanatory factor for the current crisis. *Levels marked using Generic Level Descriptors for 20m H2 essays.*	